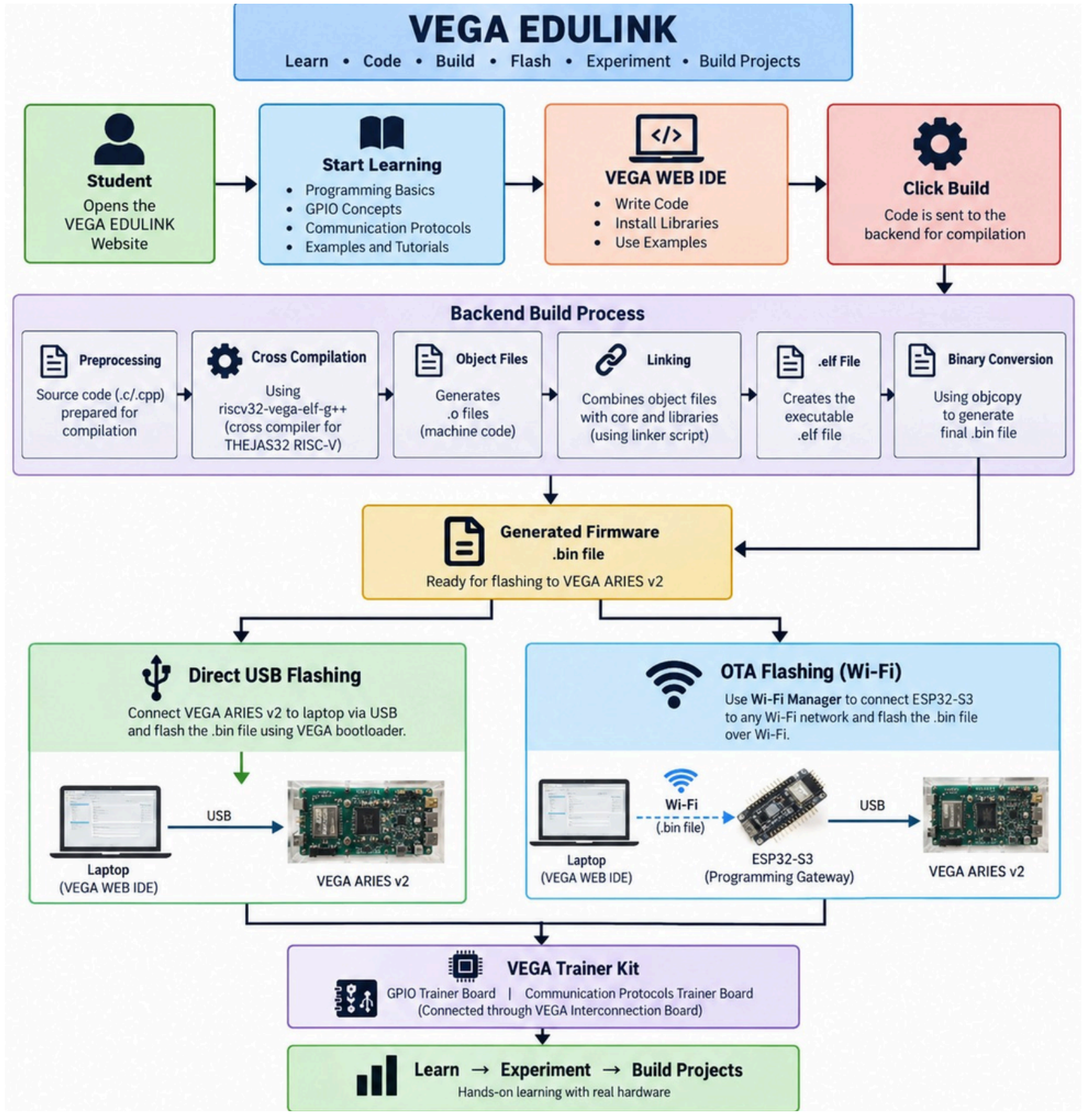
VEGA EDULINK

Learn • Code • Build • Flash • Experiment • Build

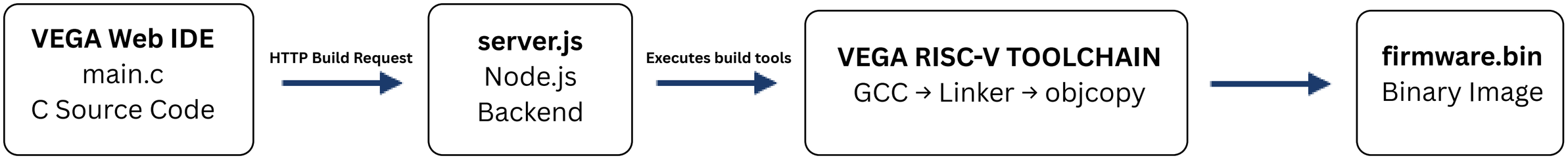
INTRODUCTION

Learning a new microcontroller can be difficult for beginners when learning materials, programming tools, libraries, examples, documentation and hardware experiments are available in **different places**. **VEGA EDULINK** was developed to bring these activities together in a **single, beginner-friendly environment for learning and developing with the VEGA ARIES v2**. The project combines the **VEGA WEB IDE with a hands-on VEGA Trainer Kit**. Together, they create a continuous path from learning basic concepts to writing programs, generating firmware, programming the board and testing the result on real hardware.

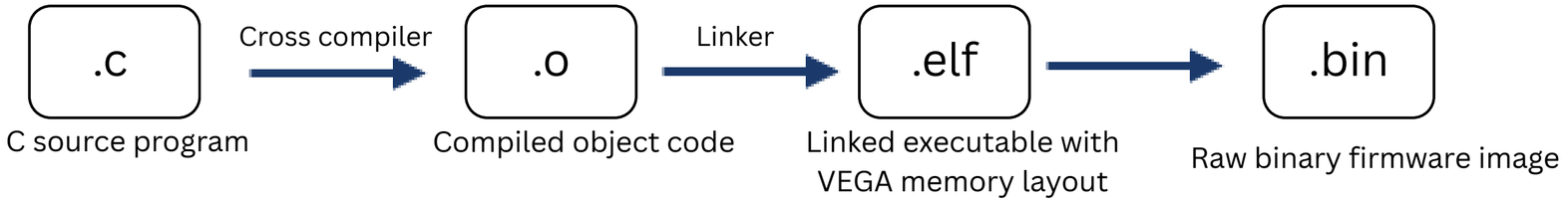


VEGA Web IDE – C to .bin Build Process

Web-based development environment and local RISC-V firmware build pipeline



C Source to Binary Transformation



Technical Toolchain

riscv32-vega-elf-gcc -> riscv32-vega-elf-ld + link1.lds -> riscv32-vega-elf-objcopy -> firmware.bin

Why a Cross-Compiler?

The VEGA RISC-V compiler runs on the development computer but generates machine code specifically for the VEGA RISC-V target.

Role of server.js:

1. Receives the **.c source code** from the Web IDE through an **HTTP** request when Build is clicked.
2. Invokes the VEGA RISC-V cross-compiler with **-march=rv32im and -mabi=ilp32** to generate .o object files.
3. Links the compiled code using **riscv32-vega-elf-ld** and link1.lds to create a single .elf file.
4. Converts the .elf executable into firmware.bin using **riscv32-vega-elf-objcopy.**
5. Controls the complete build flow in the **Node.js environment** and returns the build status/output to the Web IDE.

OTA-BASED WIRELESS PROGRAMMING FOR -VEGA ARIES V2

ESP32-S3 as a Wireless Programming Gateway :

WHAT IS OTA?

OTA — Over-The-Air
Firmware is delivered to a device over a wireless network instead of a physical programming cable.

FOTA — Firmware Over-The-Air
When OTA specifically delivers firmware updates, it's commonly called FOTA.

OTA IS AN ESTABLISHED TECHNOLOGY

Renesas | STMicroelectronics | NXP

Renesas: select RX, RL78, RA devices

STMicroelectronics: STM32 wireless family, e.g. STM32WB

NXP: automotive & connected-device FOTA solutions

OTA/FOTA is already used in industrial, IoT & automotive systems — our contribution is adapting the concept to the VEGA ARIES V2 ecosystem.

WHY OTA IN OUR PROJECT?

Conventional: Laptop → Cable → VEGA

Our approach: Web IDE → Wi-Fi → ESP32-S3 → VEGA

- Reduces repeated physical cable connections
- Convenient for programming multiple boards
- Connects the Web IDE directly to the physical VEGA hardware

Cable + OTA — Both Supported

HOW THE FIRMWARE REACHES THE GATEWAY

WEB IDE → **Build VEGA Program** → **Generate .bin** → **WiFi /HTTP Upload**



ESP32-S3 = Wireless Programming Gateway

ESP32-S3 RESPONSIBILITIES

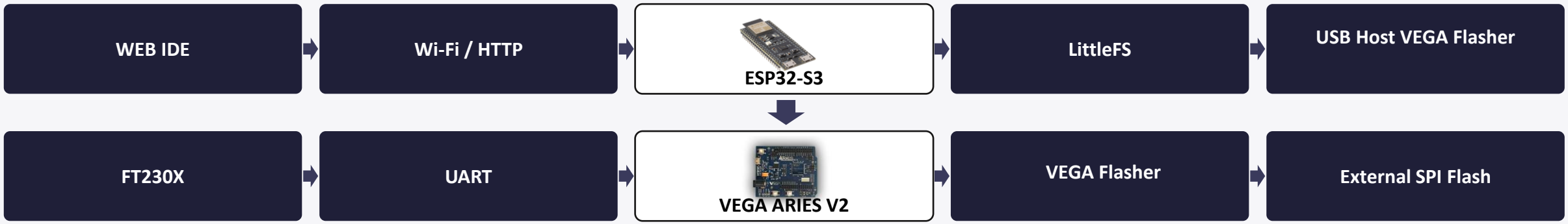
Wi-Fi → **HTTP Server** → **LittleFS** → **USB Host** → **VEGA Programming**

Why ESP32-S3?

Wi-Fi / HTTP → Receive & Store .bin → Native USB Host → FT230X → UART → VEGA ARIES V2

Dual USB interface: 1. USB-to-UART for ESP32 programming & debugging | 2. Native USB for VEGA communication

ESP32-S3 → VEGA PROGRAMMING PROCESS :



1

Web IDE → ESP32-S3

Student builds program → .bin generated → HTTP upload over Wi-Fi (HTTP POST /upload).

2

Receive & Store

Received .bin is stored in LittleFS as intermediate storage. Verified by file size + checksum/SHA-256.

3

ESP32-S3 as USB Host

Native USB port operates as USB Host, communicating with the FT230X USB-to-UART bridge.

4

FT230X → VEGA

FT230X converts USB → UART, linking to the VEGA ARIES V2 programming interface.

5

XMODEM-CRC Transfer

128-byte blocks → CRC check → ACK/NAK → retransmission on error, for reliable delivery.

STEP 6 — VEGA FLASHER: three different things, not to be confused

ESP32-S3 Firmware

Controls Wi-Fi, HTTP, LittleFS, USB Host, XMODEM & programming logic.

VEGA Flasher

Programming utility inside ESP32-S3 firmware, loaded temporarily into VEGA RAM and executed via the VEGA bootloader.

Application .bin

Student's VEGA firmware — ultimately written to VEGA external SPI Flash.

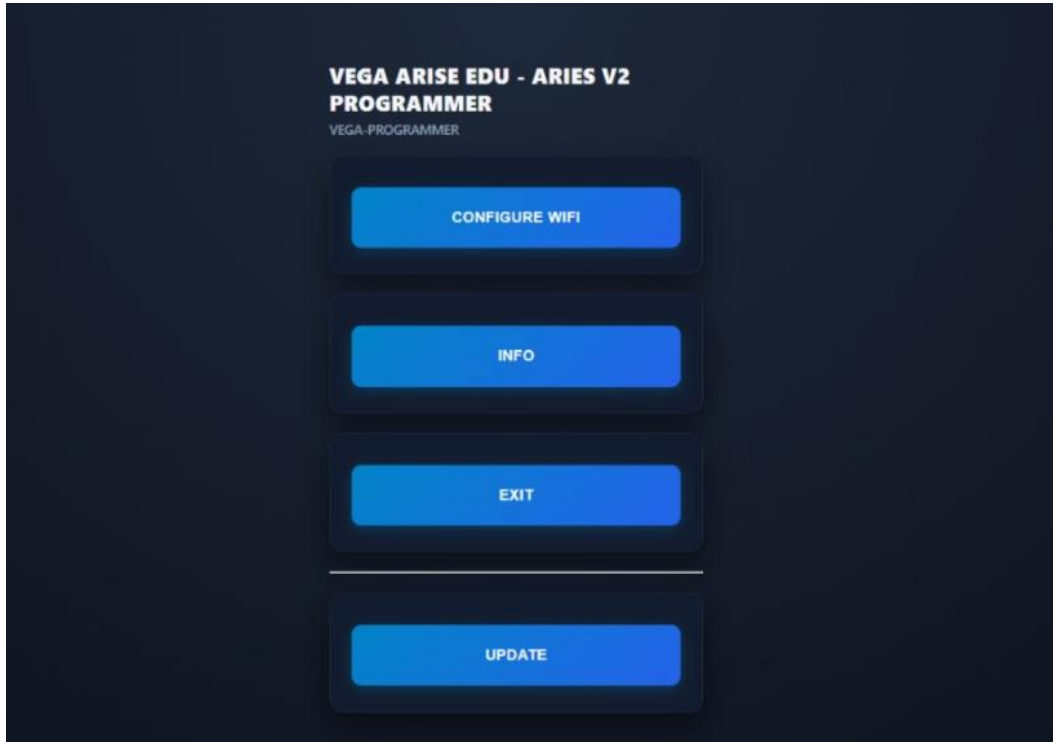
Wi-Fi Manager No saved credentials → ESP32-S3 creates AP → Configuration Portal → user selects Wi-Fi & enters password → credentials stored → automatic reconnection.

Cable + OTA Web IDE branches both ways — Cable: direct USB/UART connection to VEGA. OTA: Wi-Fi → ESP32-S3 → VEGA. Cable remains fully supported for development & debugging.

Wi-Fi MANAGER & OTA HARDWARE CONNECTION

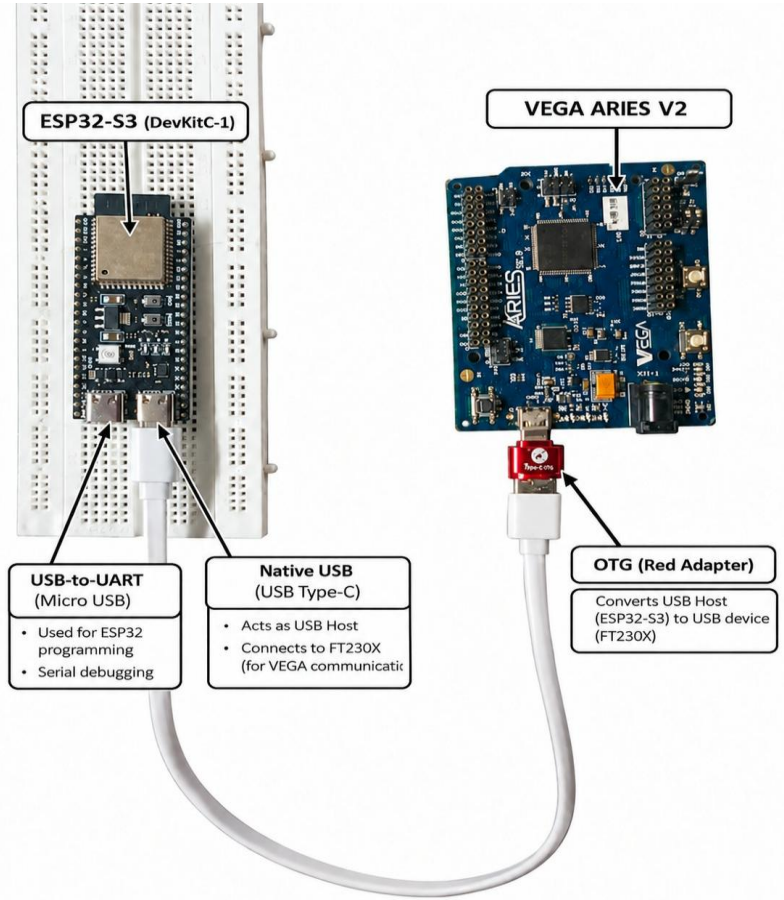
Reference views — configuration portal and physical OTA wiring :

WI-FI MANAGER — CONFIGURATION PORTAL

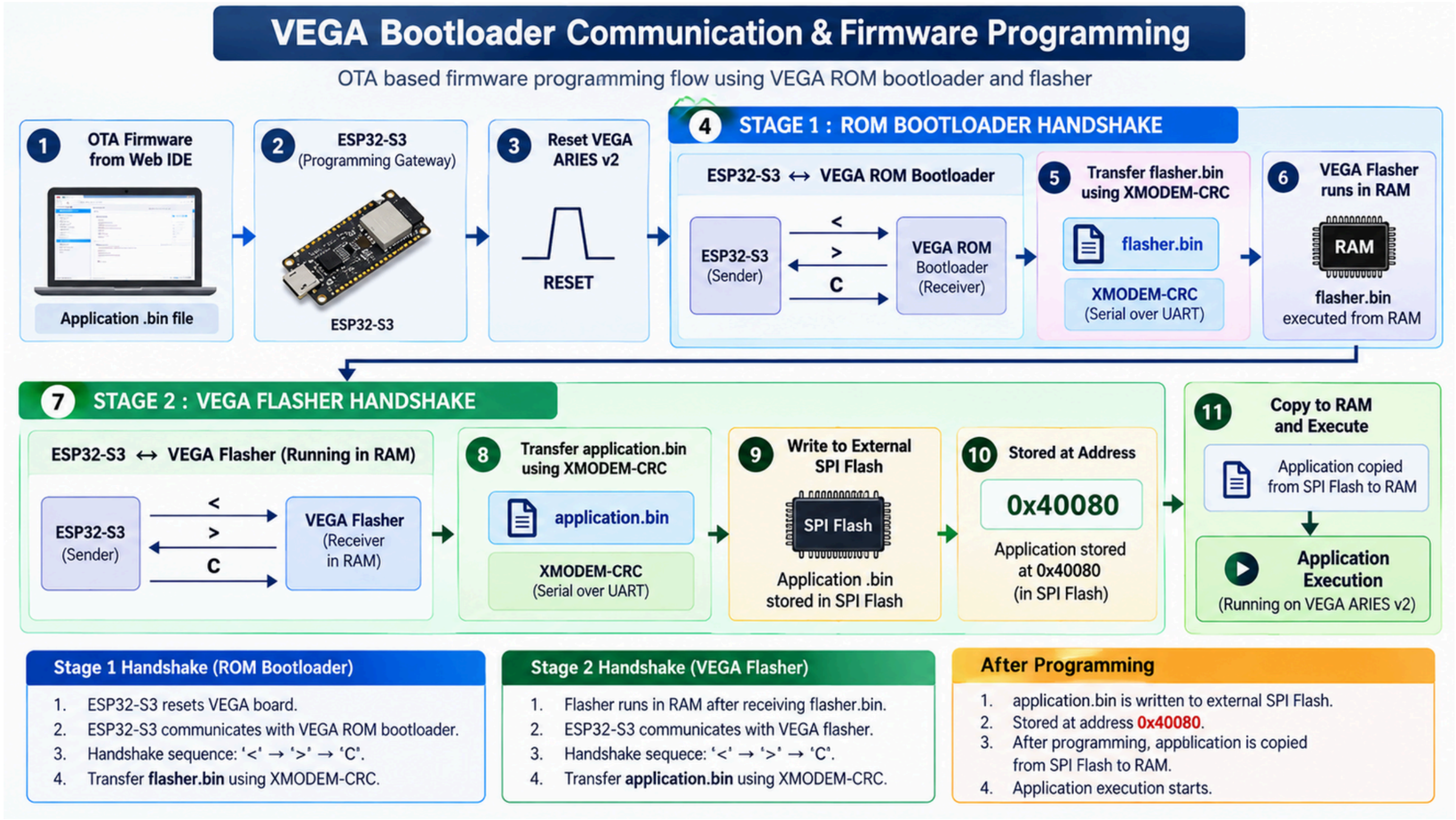


No saved credentials → ESP32-S3 creates AP → Configuration Portal (VEGA-PROGRAMMER, shown above) → user selects Wi-Fi network & enters password → credentials stored → automatic reconnection.

ESP32-S3 ↔ VEGA ARIES V2 — OTA CONNECTION

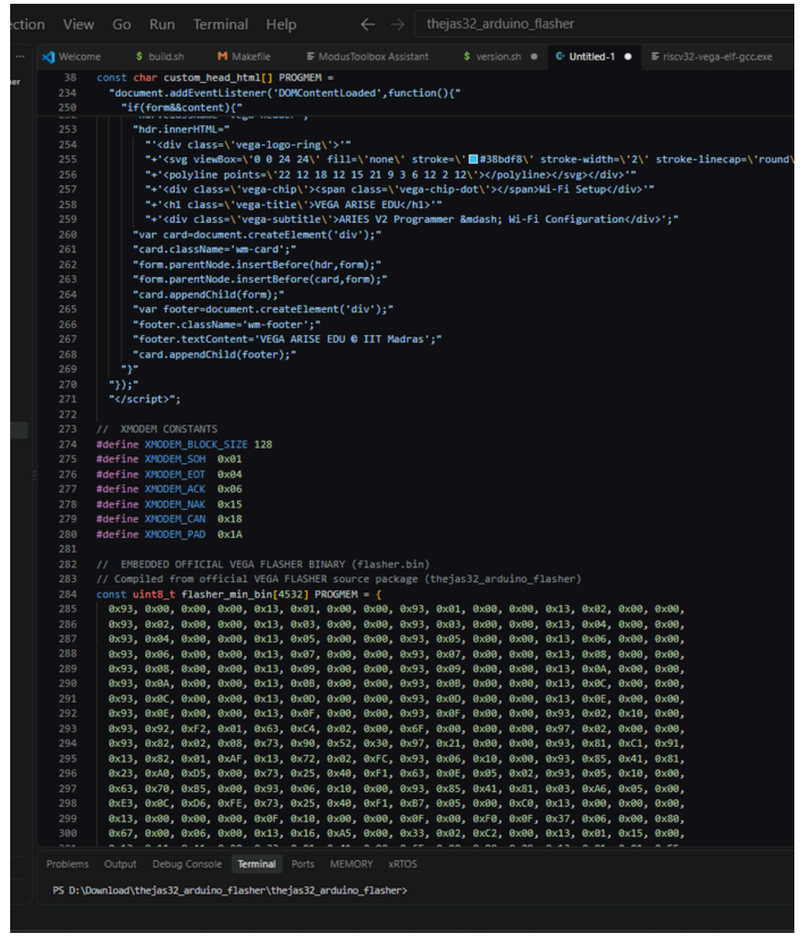


Native USB (Type-C) on the ESP32-S3 acts as USB Host and, through the OTG adapter, connects to the FT230X on VEGA ARIES V2. The separate USB-to-UART (Micro USB) port is used only for programming/debugging the ESP32-S3 itself.



Embedded VEGA Flasher

flasher.bin is embedded directly into the ESP32-S3 firmware as a byte array (flasher_min_bin[]).



Implementation

- Flasher binary stored in PROGMEM
- XMODEM-CRC parameters defined in the ESP32-S3 firmware
- Embedded flasher is transferred to the VEGA ROM Bootloader
- Transfer is performed using XMODEM-CRC

VEGA TRAINER KIT :

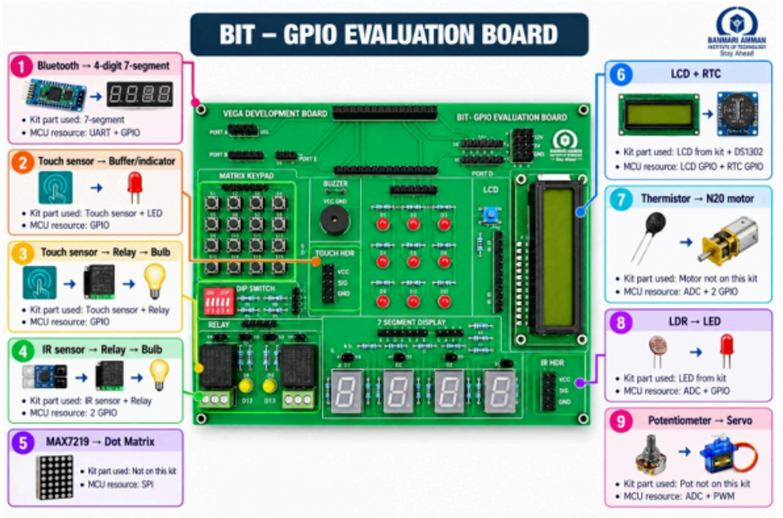
A practical hardware platform based on the VEGA ARIES v2 processor for hands-on learning and experimentation in embedded systems.

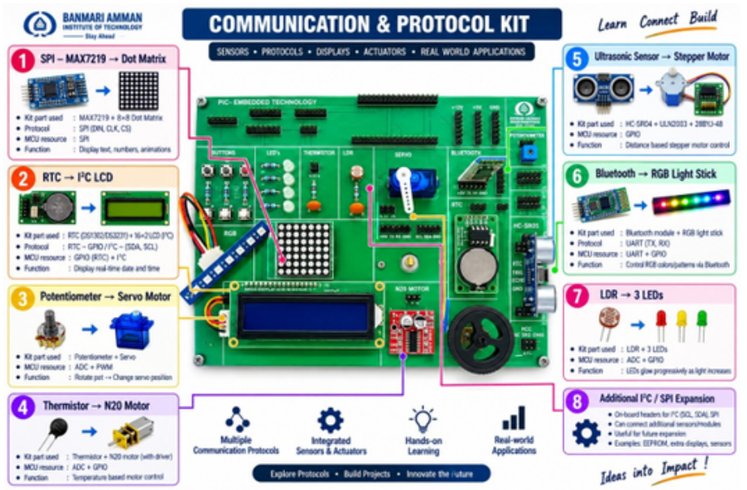
1. GPIO Trainer Board

- Provides access to VEGA ARIES v2 GPIO pins.
- Supports LEDs, switches, and digital logic-level interfacing.
- Enables practical implementation of GPIO configuration and digital I/O control.
-

2. Communication Protocols Trainer Board

- Provides dedicated interfaces for UART, SPI, and I²C.
- Supports serial data transmission, peripheral interfacing, and communication testing.
- Enables practical understanding of embedded communication protocols and hardware debugging.





FUTURE DEVELOPMENT :

Development of a dedicated VEGA Trainer Kit integrated with the VEGA EDULINK IDE, enabling learners to write, compile, flash, and debug embedded applications directly on the hardware.

- Integrated GPIO and communication peripherals for hands-on experimentation.
- Modular architecture for easy hardware and software expansion.
- Designed to support future peripheral additions and advanced embedded applications.

FUTURE UPGRADATION

01 — Enhanced VEGA Web IDE

- Integrate Serial Monitor directly into the Web IDE
- Support Serial Monitor for Direct USB & OTA programming
- Provide real-time UART output for monitoring and debugging
- Improve the workflow: Write → Build → Flash → Monitor

02 — Complete VEGA Board Package Support

- Integrate all supported VEGA board packages
- Provide board selection directly from the IDE
- Support board-specific compiler, libraries & build configurations
- Expand Web IDE support beyond ARIES v2

03 — Dedicated VEGA IDE Software

- Develop a standalone VEGA IDE desktop application
- Integrate Web IDE features into dedicated software
- Include compiler toolchain & board packages
- USB + OTA + Serial Monitor in one application
- Complete software ecosystem for VEGA development

04 — Dedicated VEGA Trainer Kit

- Integrate the VEGA Trainer Kit with VEGA EDULINK
- Enable Write → Build → Flash → Debug directly on hardware
- Include GPIO peripherals & communication interfaces
- Support hands-on embedded experimentation
- Modular design for future hardware & software expansion

05 — Cloud-Based VEGA Programming & Device Management

- Develop a **Public VEGA Server** for centralized VEGA device management
- Assign a **unique Device ID** to each VEGA ARIES V2 board for remote identification
- Enable **Internet-based firmware delivery and programming** without requiring the laptop and VEGA board to be on the same network
- Utilize the ***NINA-W10*** as the wireless programming interface between the server and VEGA ARIES V2
- Provide **real-time device status, programming progress, results and logs** through the VEGA Web IDE